Title page

Does starting age predict who stays? Competition debut and long-term participation in youth track and field

Running head: Debut age and retention in youth athletics

Author: Atle G. Guttormsen[a] (ORCID 0000-0003-0188-8462)

[a]School of Economics and Business, Norwegian University of Life Sciences (NMBU), P.O. Box 5003, 1432 Ås, Norway

Corresponding author: Atle G. Guttormsen, School of Economics and Business, Norwegian University of Life Sciences (NMBU), P.O. Box 5003, 1432 Ås, Norway. E-mail: atle.guttormsen@nmbu.no

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Ethics: The study analyses competition results registered by the Norwegian Athletics Federation and is not medical or health research as defined by Norway's Health Research Act (Helseforskningsloven §§ 2 and 4), so approval by a Regional Committee for Medical and Health

Research Ethics was not required. Sikt (the Norwegian Agency for Shared Services in Education and Research) confirmed that the use of the data raises no data-protection concerns. No participants were contacted or recruited, and informed consent was not required. No individual athlete is identified.

Author contributions: AGG conceived the study, performed all analyses, and wrote the manuscript.

Related work: A longitudinal performance-development study of the same registry is published in this journal (*J Sci Med Sport* 2026, doi:10.1016/j.jsams.2026.07.006); it concerns performance trajectories, not retention, and is cited in the Introduction. Three further studies of the registry are cited in the manuscript: a relative-age study in press (*Sports Medicine – Open*), a competition-volume cohort study under review (*International Journal of Sports Science & Coaching*), and a performance-progression study under review (*Psychology of Sport and Exercise*). They use different exposures and samples, and no text or result overlaps with the present manuscript.

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Data availability: The competition records used in this study are publicly accessible via the Norwegian Athletics Federation's online register for ages 13 and above; records for the 10–12 age classes are not publicly released by the federation. The derived analysis dataset cannot be publicly shared because it contains personal data (athlete dates of birth) under the EU General Data Protection Regulation (GDPR). The full analysis code, with the time-stamped analysis plan, will be deposited in a public repository upon acceptance. Supplementary Tables S1–S5 accompany the submission.

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